

STEP Dashboard Installation Guide

Installer handout for `install-dashboard.sh` — Ubuntu appliance setup

Overview

The `install-dashboard.sh` script transforms a fresh **Ubuntu server** (22.04 or similar) into a **STEP Dashboard appliance**: a network monitoring server built around **LibreNMS**, with supporting tools and a customer-facing setup wizard.

Before You Start

Requirement	Details
Operating system	Ubuntu (script targets 22.04; PHP 8.3 paths work on newer releases)
Privileges	Root access — run under <code>sudo bash</code>
Repository on disk	Project must exist at <code>/opt/dashboard-repo</code> (STEP NetTools, syslog schema, web installer files)
Network	Internet access for APT, Git clone, and downloads
Duration	Allow sufficient time — package installs, Composer, and DB migrations take a while

Important: The script stops on any error (`set -e`). If a step fails, the install halts rather than continuing silently.

How to Run

```
sudo bash
bash /path/to/ref_image_build/install-dashboard.sh
```

Information You Will Be Asked For

- Web server hostname** — Used for Nginx and the LibreNMS URL (e.g. `dashboard.customer.local`)
- MariaDB database password** — For the `librenms` MySQL user and LibreNMS `.env`
- LibreNMS system user password** — For the Linux `librenms` service account

4. **Web admin password** — For LibreNMS login user `admin` (must be strong or creation may fail)
5. **Timezone** — US timezone menu (Eastern through Guam; default is Eastern)

What the Script Does

1 Base Packages

Updates APT and installs dependencies including:

- **Web:** Nginx, PHP 8.3 FPM and extensions
- **Database:** MariaDB
- **Monitoring:** SNMP, RRDtool, fping, nmap, mtr, and related tools
- **Utilities:** git, curl, cron, vim, tcpdump, Python packages, and more

2 LibreNMS Application

- Creates the `librenms` Linux user (home: `/opt/librenms`)
- Clones LibreNMS from `https://github.com/sk12dev/librenms.git` (custom fork)
- Sets directory permissions and ACLs on `rrd` , `logs` , `storage` , and `cache`
- Installs PHP dependencies via Composer
- Configures PHP and system timezone

3 MariaDB

- Tunes MariaDB settings and restarts the service
- Creates database `librenms` (utf8mb4)
- Creates user `librenms@localhost` with your chosen password

4 Web Stack (Nginx + PHP-FPM)

- Creates a dedicated PHP-FPM pool for LibreNMS on `/run/php-fpm-librenms.sock`
- Configures Nginx at `/etc/nginx/conf.d/librenms.conf`
- Serves LibreNMS at `/` on port 80
- Serves STEP NetTools at `/step-tools/`
- Removes the default Nginx site and restarts services

5 SNMP & Scheduling

- Configures SNMP agent (community set to `public` — change for production)
- Installs LibreNMS cron jobs and systemd scheduler timer
- Sets up log rotation and the `lnms` CLI command

6 LibreNMS Configuration & Users

- Writes `/opt/librenms/.env` with database credentials and `APP_URL`

- Generates Laravel application key and runs database migrations
- Creates `admin` user (admin role) — your web login
- Creates `api-ro` user (global-read) for STEP NetTools API access
- Generates an API token for `api-ro` — **shown once at end of install**
- Marks installation complete (skips LibreNMS web installer)
- Runs validation check and fixes log file ownership

7 Syslog-ng

- Installs syslog-ng package
- Creates the `ilog` database and `syslog` table for storing syslog in MySQL
- Restarts syslog-ng

Note: Copying STEP-specific syslog-ng routing configs is currently commented out in the script. Only the package and database are set up. Full syslog routing may require a follow-up step.

8 LibrePulse

Downloads the LibrePulse heartbeat script to `/opt/librepulse/` for remote appliance monitoring.

9 STEP NetTools

- Deploys web UI to `/opt/step-tools/`
- Writes `/etc/step-tools/config.php` with LibreNMS URL and API token
- Installs `generate-api-key.sh` for regenerating API tokens later

STEP NetTools provides Device Inventory, Alerts, Ports, and Speedtest — all powered by the LibreNMS API.

10 Customer Web Installer

- Deploys setup wizard to `/opt/customer-web-installer/`
- Installs sudoers rules for network configuration scripts
- Configures Nginx on **port 8080** as the default site for the wizard

The wizard allows end customers to:

- **Step 1:** Configure static IP (interface, IP, gateway, subnet, DNS) via Netplan
- **Step 2:** Run a lighter post-setup script — not a re-run of this full installer

When Installation Completes

Component	Access
LibreNMS	<code>http://<hostname>/</code> — user: <code>admin</code>
LibreNMS API	User: <code>api-ro</code> — token printed once at end of install
STEP NetTools	<code>http://<hostname>/step-tools/</code>
Customer setup wizard	<code>http://<server-ip>:8080/</code>

Installer Checklist

1. Confirm Ubuntu server has network and root/sudo access
2. Ensure `/opt/dashboard-repo` contains this project
3. Run `install-dashboard.sh` and answer all prompts
4. **Copy the API token immediately** when displayed
5. Verify LibreNMS loads at the hostname URL
6. Verify STEP NetTools loads at `/step-tools/`
7. Share customer wizard URL (`http://<ip>:8080/`) if needed for network setup
8. Change default SNMP community (`public`) for production environments

Important Notes

- **API token:** Copy it when shown — it will not be displayed again. Run `generate-api-key.sh` if missed.
- **Two install paths:** This script is the full install. The customer wizard's "Start Installation" runs a smaller script (`configure-dashboard-from-web.sh`), not this full installer.
- **Re-runs:** User creation skips if `admin` or `api-ro` already exist. API token generation skips if a token already exists.
- **Syslog-ng:** Database is created; custom routing configs are not yet applied automatically.

Architecture Overview

Ubuntu Server | ▼ `install-dashboard.sh` | — LibreNMS (monitoring core) — MariaDB (librenms + iolog databases) — Nginx + PHP-FPM (web front end) — SNMP + cron + scheduler (polling & maintenance) — Syslog-ng (package + iolog DB) — LibrePulse (heartbeat) — STEP NetTools (customer-facing tools UI) — Customer Web Installer (port 8080 – network setup wizard)

Troubleshooting

Issue	What to check
Wizard page won't load after network change	Netplan apply changed the IP — browse the new IP shown by the wizard
Install appears stuck in wizard step 2	Check <code>/tmp/install.log</code> on the appliance
STEP NetTools can't reach LibreNMS	Verify <code>/etc/step-tools/config.php</code> has valid URL and API token
API token not generated	Run <code>generate-api-key.sh</code> as root
Validation warnings at end	Review output from <code>validate.php</code> — install continues but issues may need attention